IDC® - Bearings
by INTERPRECISE Germany

Tandem bearings



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Interprecise Donath GmbH with it's brand IDC® stands for flexibility and innovation, customer oriented thinking and quality focused work.
It is our philosophy to realize economical solutions for demanding customer-requests through technological know how and state of the art production equipment.

The company was founded in 1980 and is situated in the technological cluster of Herzogenaurach, well known for it's tradition and competence in bearing manufacturing.

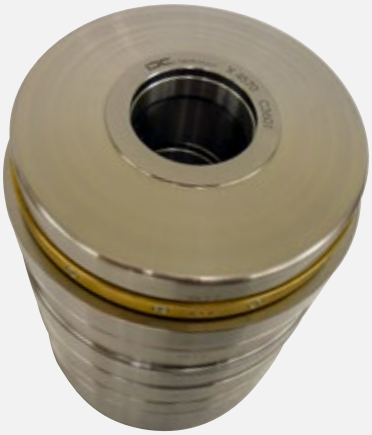




Interprecise Donath GmbH offers a vast assortment of standard bearings based on DIN-ISO, as well as multiple variations thereof and customer specific special bearings.

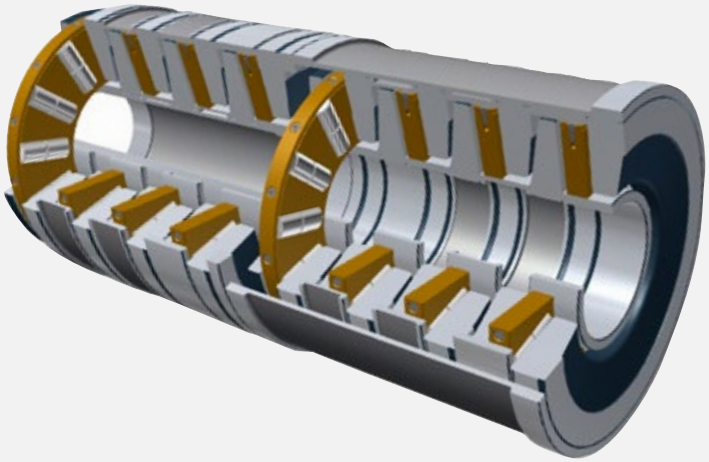
Upon customer request we manufacture bearings in higher accuracy classes, with specific internal clearance, with greases selected for the application, individual materials for bearing seals, hybrid bearings as well as various surface finishes and coatings.

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Extruder gearboxes with their requirements for high axial load capabilities in a narrow envelope, paired with the expectation of high reliability and long service life, are an ideal application for products and solutions by IDC.

To bear extreme thrust loads in a narrow envelope, multilayer thrust cylindrical roller bearings are used. Several, stacked, thrust cylindrical roller bearings are called tandem bearings. Mostly tandem bearings are used in gearboxes for double screw extruders. Depending on the design concept two tandem bearings or one tandem bearing paired with a standard thrust cylindrical roller bearing are being used.



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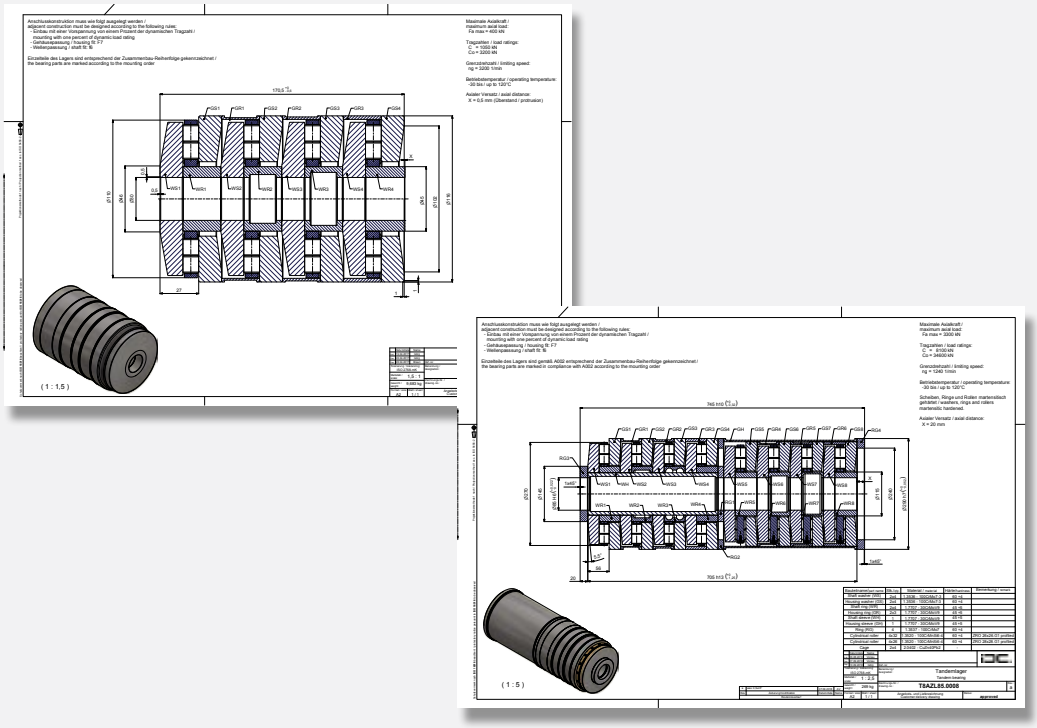
To distribute the loads and to avoid only one or two bearings having to carry the full thrust, the stages/levels are separated by elastic sleeves. Elastic sleeves and washers have to be perfectly matched to achieve an even distribution over every stage.

Shaft washers and housing washers are designed to suffer an almost identical elastic deformation under load, to keep the roller raceways parallel.

Designing and calculating the sleeves for bore and housing is critical for an optimum function of the bearing. Shape and dimensions influence their spring characteristics. Spring characteristics of the rings should be even over the full stretch of permissible axial loads and not only for the working point.

Most tandem bearings have three, four, six or eight stages. On request we will design your specific configuration for your specific needs.

Interprecise produces tandem bearings starting with an inner diameter of 19mm.



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Research and development traditionally play an important role at Interprecise.

During the design process all critical parameters of the bearing, concerning function, customer value and reliability of the product are defined, thus determining all further costs for the product life-cycle.

A thorough communication with our customers ensures that product requirements, application conditions and customer objectives are well understood and precisely described. Our highly qualified engineers and technicians analyse the information.

Concepts are drafted with consideration for maximum customer benefit and economical aspects while meeting the product requirements.



Drafts and designs are made by our skilled employees with a 3D-design program. Results of the design-process are verified and evaluated through methods of finite elements, multi- body-simulation and analytical calculation methods. They are validated through prototyping and test-samples.

Changes during the product development can be traced through our product-management-system at any time.

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To achieve a high level of quality, flexibility and efficiency, we execute the main operations turning and grinding internally.

To achieve our objectives we work with modern state of the art CNC equipment.

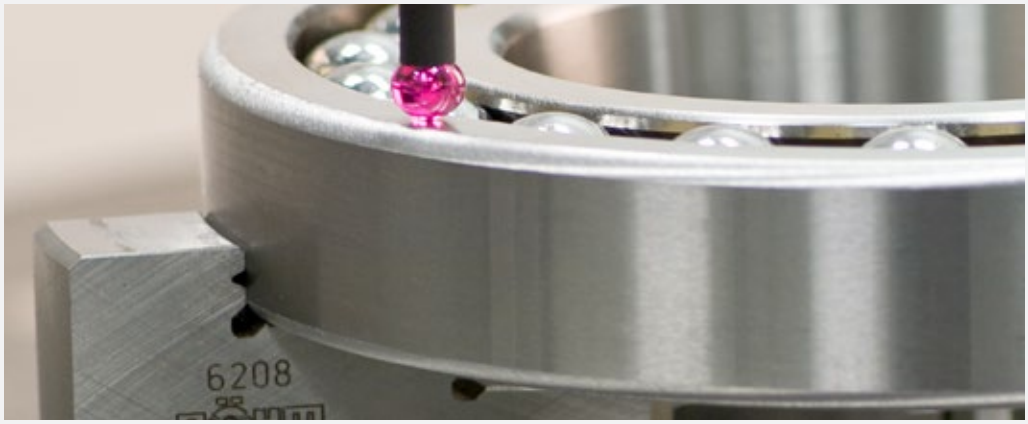


All of our activities in the company are subject to our commitment to quality.

Quality assurance is conducted in each stage and step of the product development.

Our employees use the latest measuring equipment available in the industry.

All process' of the company are certified to ISO 9001:2015

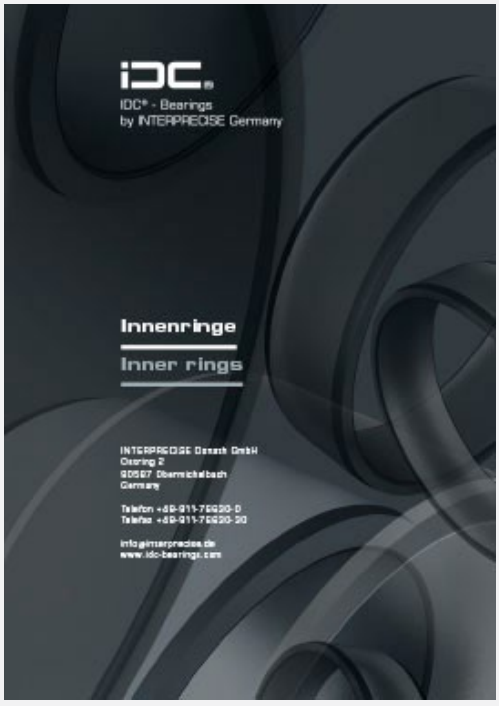


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Further product groups of IDC®









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